Lesson plan · for the teacher

Finding groups

Data science · Level 4 · Lesson 2

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Tell clustering from classification.
- Run k-means steps.
- Use groups with care.

Activity: Human k-means

1. Warm-up (5 min): Show an unlabelled scatter plot of library loans and days kept. Ask colleagues to circle any groups they see.
2. Teach (8 min): Do one k-means round on the board with two centres and six dots.
3. Run (15 min): Groups run k-means on a small anonymous table for k = 2, then k = 3.
4. Act (7 min): Agree on neutral names and one useful action for each group.
5. Wrap-up (5 min): Ask two colleagues to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Did k = 2 or k = 3 help more?
- Which group name could harm a pupil?
- What will you do for each group?

Exit ticket

- Centres at 3 and 10. Which centre does a pupil with 6 loans join?

Try it at school tomorrow
Plot pupils' sleep and screen time without names, then let three pupils stand as centres. Everyone joins the nearest, and the class works out where each centre moves.

Practice questions · for you

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1 **Does clustering need examples that already have labels?**

◯ No, it finds groups on its own ◯ Only for k = 1 ◯ Yes, always

2 **Put the k-means steps in order.**

☐ Move each centre to its mean

☐ Choose k and place centres

☐ Repeat until nothing changes

☐ Each case joins its nearest centre

3 **Do one k-means step: each pupil joins the nearer centre, then print the two new centres.**

```
loans = [1, 2, 3, 8, 9, 10]
c = [2, 5]
g0 = []
g1 = []
print(c[0], c[1])
```

Write your program:

4 **Clusters of library habits were found. Which use is best?**

◯ Ranking pupils by cluster ◯ Choosing books and invitations to a reading club

◯ Writing “lazy reader” on report cards

5 **k-means always finds the same groups, whatever the starting centres.**

◯ True ◯ False

✎ **Which groups of pupils does your school already talk about? Do those names help or harm?**

Check your answers

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1 **No, it finds groups on its own**

Clustering works without any labels.

2 **Top to bottom: 3, 1, 4, 2**

Assign, move, repeat: the two middle steps take turns until the groups settle.

3

```
loans = [1, 2, 3, 8, 9, 10]
c = [2, 5]
g0 = []
g1 = []
for x in loans:
    if abs(x - c[0]) <= abs(x - c[1]):
        g0.append(x)
    else:
        g1.append(x)
print(sum(g0) / len(g0), sum(g1) / len(g1))
```

1, 2 and 3 join the centre at 2; 8, 9 and 10 join the one at 5.

4 **Choosing books and invitations to a reading club**

Clusters are good for planning support, not for labelling children.

5 **False**

Different starts or a different k can give different groups, so try a few.